



Do extreme range estimators improve realized volatility forecasts? Evidence from G7 Stock Markets

Burak Korkusuz, Dimos Kambouroudis, David G. McMillan^{*}

Division of Accounting and Finance, University of Stirling, UK

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ABSTRACT

This paper investigates whether range estimators contain important information in forecasting future realized volatility. We use widely applied range-based estimators: Parkinson, Garman-Klass, Roger-Satchell, and Yang-Zhang within a HAR-RV-X framework. Overnight volatility and close-to-close volatility estimators are also included, and the forecasting exercise is applied to G7 stock markets using a rolling window. Using QLIKE, HMSE and MCS forecast criteria, several noteworthy points are reported. The overall findings suggest that while no single model dominates, overnight return volatility achieves the most consistent performance. For example, HAR-RV model forecasts for CAC and DAX indices are improved only by overnight volatility, with some evidence also for SPX. For other indices, forecasts are improved by Parkinson and/or Garman-Klass volatility estimators. Of note, simpler range estimators outperform more complex range estimators. The findings could be important for investors in managing portfolio risk.

1. Introduction

In modeling and forecasting stock market volatility, the dominant approach uses the realized volatility (RV, Andersen and Bollerslev, 1998) measure, modelled with the HAR-RV specification of Corsi (2009).¹ The subsequent literature (e.g., Degiannakis and Filis, 2017; Kambouroudis et al., 2021; Peng et al., 2018; Liu et al., 2019; Wang, 2019; Yang and Liu, 2012; Zhang et al., 2020) seeks to improve forecast accuracy by adding exogenous variables in a HAR-RV-X model. These exogenous variables include, for example, cross-market information, implied volatility and EPU (economic policy uncertainty). Further explanatory variables such as the leverage effect, realized semi-variance, jump, and overnight volatility are considered as more directly related to the dependent variable as they capture the stylised facts of volatility. For example, Barndorff-Nielsen et al. (2010) introduce positive and negative realized semi-variance, Corsi and Reno (2012) consider leverage, while Wang et al. (2015) also examine the role of overnight volatility.

An alternative source of information that can improve forecasts is given by OHLC (Open, High, Low, Close) prices. Although RV is the sum of squared intraday returns, it does not necessarily capture high-low range information. Thus, range estimators (e.g., Parkinson, Garman-Klass, Roger-Satchell, and Yang-Zhang) can be introduced to the HAR-RV-X framework. Range estimators (or OHLC estimators) are straightforward to calculate given the availability of open, high, low and close price data. Despite both RV and OHLC being volatility estimators, they capture different frequencies in the data and the information value of OHLC estimators for RV-based forecasts has not been examined. Therefore, this study fills the gap in this literature by investigating whether OHLC estimators that

^{*} Corresponding author at: Accounting and Finance Division, University of Stirling, FK9 4LA, UK

E-mail address: david.mcmillan@stir.ac.uk (D.G. McMillan).

¹ Andersen et al. (2001) indicate that Realized Variance (RV) is a natural estimator for the integrated variance.

embed range information improve future RV forecasts.

Identifying our work from the existing literature, we can consider, for example, Peng et al. (2018) and Todorova and Husmann (2012). While both focus on the forecasts for a single market, the former incorporates 'X' variables into the RV equation and the latter uses the range-based measures as the series to be forecast (i.e., the proxy for volatility). A further paper, Kambouroudis et al. (2021) does include a number of markets in their forecast exercise, but does not include range-based 'X' variables.² In recent work that considers range-based forecasts, Petneházi and Gáll (2019) combine with a neural network approach for the Dow Jones Industrial Average, while Wu and Xu (2022) apply the CARR approach to Chinese stock markets. Our paper, in essence, combines these approaches by allowing the range-based measures to be the 'X' variables across a selection of markets.³ In doing so, we believe this paper addresses a clear gap in the literature.

2. Empirical methodology

2.1. Realized variance

Volatility is latent and therefore requires a proxy measure. The use of squared (or absolute) returns form a traditional, but noisy, approach that are largely superseded by realized variance. This latter measure is based on high frequency data, thus containing more information, and considered to be a better representation of true volatility (Andersen and Bollerslev, 1998; Andersen et al., 2003) and is given by:

$$RV_t = \sum_{i=1}^m r_{t,i}^2 \quad (1)$$

where realized variance (RV) is calculated as the sum of squared intraday returns (r), with m the number of intraday observations during day t . Martens (2001) finds that a higher m provides a more accurate daily volatility estimate. However, if m is too high, the efficiency of high frequency data can be distorted by microstructure effects. Hol and Koopman (2002) suggest selecting a frequency interval between 5 and 30 min, while Liu et al. (2015) compare over 400 different realized measures and argue that 5-minute RV is preferred.

2.2. HAR-RV-X model

Corsi (2009) proposes the HAR-RV model as an additive cascade model of different volatility components and is given by:

$$RV_{t+h}^d = \beta_0 + \beta_d RV_t^d + \beta_w RV_t^w + \beta_m RV_t^m + \beta_X X_t^d + \varepsilon_{t+h} \quad (2)$$

where RV_t^d is the daily volatility component; RV_t^w the weekly component, and RV_t^m the monthly component. These latter two components are given by:

$$RV_t^w = \frac{1}{5} (RV_{t-5}^d + RV_{t-4}^d + \dots + RV_{t-1}^d) \quad (3)$$

$$RV_t^m = \frac{1}{22} (RV_{t-22}^d + RV_{t-21}^d + \dots + RV_{t-1}^d) \quad (4)$$

An advantage of the HAR-RV model is that it can be estimated using OLS, while the model parameters are designed to represent short-, middle-, and long-term investors, the reactions of which drive volatility. In Eq. (2), the $\beta_X X_t^d$ refers to any included exogenous variables.

2.3. Range based volatility estimators

We consider several range-based volatility estimators derived from OHLC prices. Close-to-close volatility is presented in Eq. (5) as:

$$V_{CC,t} = \left(\ln \frac{C_t}{C_{t-1}} \right)^2 \quad (5)$$

Close-to-close volatility is simply squared daily returns and is one of the most widely used volatility measures and often employed as the true volatility proxy. Close-to-close volatility, in capturing only closing prices, is perhaps of most interest to long-term or passive investors. However, investors who focus on intraday trading are also interested in OHLC prices. Eq. (6) presents the overnight volatility

² Another approach is the CARR (conditional autoregressive range) model of Chou (2005), see Xie (2018) for a recent application.

³ A further approach, suggested by Christensen and Podolskij (2007) and Martens and van Dijk (2007) also combines the RV and range approaches and does so by calculating the range within intraday intervals before summing to the daily frequency. This requires the availability of intraday interval high and low data as opposed to daily high and low as used here. This would be an interesting avenue to pursue, however, we currently lack availability of the required data.

estimator proposed by Brooks et al. (2000).

$$V_{ONV,t} = \left(\ln \left(\frac{O_t}{C_{t-1}} \right) \right)^2 \quad (6)$$

This measure requires today's open and yesterday's close prices to capture accumulated overnight information and is argued to improve volatility forecasts (see, for example, Wang et al., 2015; Kambouroudis et al., 2021).

An alternative to open and close prices is to consider high and low prices. Parkinson (1980) proposes the scaled high-low price range to measure the variability of stock returns and is given by:

$$V_{PK,t} = \frac{1}{4\ln 2} \left(\ln \frac{H_t}{L_t} \right)^2 \quad (7)$$

Range estimators assume that the stock prices follow a geometric Brownian motion with two parameters, drift⁴ and volatility. However, the Parkinson estimator assumes zero-drift price process. Garman-Klass (GK; 1980), in Eq. (8), suggest a new estimator that incorporates the drift process, as such:

$$V_{GK,t} = 0.5 \left(\ln \frac{H_t}{L_t} \right)^2 - (2\ln 2 - 1) \left(\ln \frac{C_t}{O_t} \right)^2 \quad (8)$$

In essence, the GK estimator is a weighted average of the Parkinson volatility estimator and the drift (open-to-close squared return), although they assume as with Parkinson, that the price is a zero-drift process.

Building upon this, Rogers and Satchell (RS; 1991) propose another measure of volatility, which is given by Eq. (9):

$$V_{RS,t} = \left(\ln \frac{H_t}{C_t} \right) \left(\ln \frac{H_t}{O_t} \right) + \left(\ln \frac{L_t}{C_t} \right) \left(\ln \frac{L_t}{O_t} \right) \quad (9)$$

Rogers and Satchell (1991) address the issue of drift more formally with their estimator which is independent of drift process and thus, does not require the assumption of zero-drift.

Yang and Zhang (YZ; 2000) develop a new volatility estimator, given in Eq. (10), to further enhance the efficiency gain over previous volatility estimators. They argue their estimator is the minimum-variance unbiased estimator and independent of the drift and opening jumps of the underlying price movements.

$$V_{YZ,t} = \left(\ln \frac{O_t}{C_{t-1}} \right)^2 + k \left(\ln \frac{C_t}{O_t} \right)^2 + (1-k)V_{RS,t} \quad k = \left(\frac{0.34}{1.34 + \frac{n+1}{n-1}} \right) \quad (10)$$

Within the existing literature (e.g., Molnar, 2012) there is some comparative work of the range-based volatility estimators. The GK estimator is found to be the optimal OHLC estimator by Bali and Weinbaum (2005), Li and Hong (2011), Todorova and Husmann (2012), Jiang et al. (2014). Raju and Rangaswamy (2017) suggest the YZ estimator for in and out-of-sample forecasting performance. In contrast, Yarovaya et al. (2016) find inconclusive results. This work, however, is not conducted in the context of the HAR-RV-X model.

2.4. Rolling window forecasts and loss functions

The out-of-sample forecasts are obtained using a rolling window approach. The advantage of this technique is in allowing parameter values to change over time. However, it should be noted that window size choice is arbitrary, balancing between using too little data that can cause imprecise estimation and too much that leaves too few observations in the forecast period. We use a window of 600 observations, which covers a period of over two years.

To evaluate the out-of-sample performance of competing models, we employ two loss functions that are known to be robust in the context of heteroscedasticity, namely the quasi-Gaussian log-likelihood (QLIKE) and heteroscedasticity adjusted mean squared error (HMSE).⁵

These two measures are given by Eqs. (11) and (12):

$$QLIKE = \frac{1}{\tau} \sum_{t=T+1}^{T+\tau} \left[\frac{RV_t}{\widehat{RV}_t} - \log \left(\frac{RV_t}{\widehat{RV}_t} \right) - 1 \right] \quad (11)$$

⁴ Drift process means the difference between open and close prices during a trading day.

⁵ The robustness of the QLIKE and MSE is documented in the seminal paper of Patton (2011). Whilst assessing different forecasts for the mean, the MSE criterion could be seen as a natural choice. However, according to Bollerslev et al. (1994) and Bollerslev and Ghysels (1996), the MSE loss function may be less clear in a heteroskedastic environment. For this reason, Bollerslev and Ghysels (1996) suggest the heteroskedasticity-adjusted MSE (HMSE). Given the heteroskedastic feature of financial volatility, the HMSE is the loss function of our choice over the MSE.

$$HMSE = \frac{1}{\tau} \sum_{t=T+1}^{T+\tau} \left[1 - \frac{\widehat{RV}_t^2}{RV_t^2} \right]^2 \quad (12)$$

Where $\widehat{RV}_t$ denotes the out-of-sample forecasts from the competing models and RV_t is true volatility, τ is the number of out-of-sample periods.

2.5. Model confidence set (MCS)

Hansen et al. (2003) suggest the MCS procedure to identify a set of superior models through a specific elimination algorithm. For a given level of confidence, the elimination algorithm examines which group of models survive among a set of competing models, defined for a specific loss function without a benchmark model. Poorly performing models are eliminated from the set of competing models. Briefly, for the MCS procedure, let $L_{i,k}$ denote the criterion of model i and $d_{i,j,k} = L_{i,k} - L_{j,k}$ is the differential. The null hypothesis of MCS procedure is $H_{0,M} = E(d_{i,j,k}) = 0$, for $i, j \in M$, $M \subset M^0$ and the null is tested against the alternative $H_{1,M} = E(d_{i,j,k}) \neq 0$, for some $i, j \in M$.

3. Data

The data used in this study consists of the stock market indices of G7 (group of seven), namely Canada, France, Germany, Italy, Japan, United Kingdom, and United States. Daily data for closing prices, range prices (open, high, low, close), and 5-minute realized variance is taken over the period from June 1, 2009 to October 22, 2021. The data is obtained from the Oxford-Man Institute's Quantitative Finance Realized Library. As trading days across the G7 stock markets differ, the data cleaning process involves forming a common data sample, with non-values in any of the markets being omitted.

Table 1 presents the descriptive statistics for all series after the data alignment process and leads to approximately 2780 observations for each series. The first 600 observations (2009/06/02 – 2012/01/18) are used for in-sample estimation, with the remaining (2180) observations (2012/01/19 – 2021/10/22) used for out-of-sample one-step-ahead rolling window forecasts. The summary statistics are as expected for this data. For example, the mean of all the series is close to zero. Negative skewness is found for the returns series, with positive skewness for the volatility measures. Excess kurtosis is reported for all series, and, in turn, significant non-normality is noted. Ljung-Box Q-statistics support the presence of autocorrelation, which is stronger for the volatility series, while augmented Dickey-Fuller (ADF) tests reject the null hypothesis of a unit root.

4. Empirical results

Using one-step-ahead rolling windows, we evaluate the forecast results of seven competing models (HAR-RV as the baseline and the OHLC range-based estimators: close-to-close, overnight volatility, Parkinson, Garman-Klass, Roger-Satchell, and Yang-Zhang) for 5-minute realized volatility for seven international markets (for clarity, Table 2 summarises the models and stock markets). To evaluate the forecasts, we use the QLIKE and HMSE loss functions, which are regarded as robust (Patton, 2011), while the MCS procedure identifies the set of preferred models.⁶

The forecast results are presented in Table 3. The realized volatility forecasts for the CAC and DAX stock indices suggest that HAR-RV forecasts can be improved upon by the addition of overnight information. The range estimators (Parkinson, Garman-Klass, Roger-Satchell, and Yang-Zhang) do not provide additional extra information over the HAR-RV benchmark model. For the results of the FTSE index, the overnight variable does not improve the forecasting accuracy over the HAR-RV model, in contrast with the results for the other indices. This result, however, is consistent with the findings of Kambouroudis et al. (2021), who note the FTSE as the exception among other indices for which the overnight information does not improve forecast performance. Unlike the CAC and DAX indices, realized volatility forecasts for the FTSE are improved by the Parkinson range estimator (HAR-RV-PK). The inclusion of more sophisticated range-based estimators such as Roger-Satchell and Yang-Zhang do not improve forecasts. In the case of FTSEMIB index, QLIKE and HMSE produce different results, although a common result is that HAR-RV-ONV is the second-best model on both criteria. This supports the view that the overnight variable can be considered an important variable. The QLIKE measure supports the HAR-RV-CC and HAR-RV-ONV, while the range estimators are outperformed by the benchmark model. The HMSE criteria selects the HAR-RV-GK as the preferred model, while HAR-RV-ONV and HAR-RV-YZ are also supported.

The results for the NIKKEI index indicate that the HAR-RV-CC model outperforms other models across both loss functions. However, the ranking set of NIKKEI results is inconclusive. For example, the QLIKE also favours the HAR-RV model, while range estimators do not contain additional information. The HMSE, however, shows that the HAR-RV-CC and HAR-RV-YZ models improve the forecasting accuracy compared to the benchmark model. Similar to the NIKKEI index, the best-performing forecast models for the SPX index also differ across loss function. The QLIKE suggests HAR-RV-ONV as the best-performing forecasting model, followed by HAR-RV-PK, HAR-RV-YZ, and HAR-RV-GK models respectively. The model rankings based on HMSE support the HAR-RV-PK, HAR-RV-GK, HAR-RV-RS, and HAR-RV-YZ respectively.

Taking an overview of the results, even though no single model dominates, we can provide some general comment. The inclusion of

⁶ We also conduct recursive window forecasts to provide robustness, the results of which are available upon request.

Table 1

Descriptive statistics of G7 data (return, range, and 5-min RV series).

<u>RETURN</u>	MEAN	STD. DEV.	SKEW.	EX. KURT.	JARQUE-BERA	Q(5)	ADF
CAC	0.0002	0.0135	-0.5417	7.8268	7229.12***	19.4829***	-13.49***
GDAXI	0.0003	0.0133	-0.4563	7.9884	7485.72***	16.6816***	-27.25***
FTSE	0.0001	0.0107	-0.8529	8.4088	8524.27***	24.9361***	-13.59***
FTMIB	9.3E-05	0.0165	-1.0084	10.561	13,386.5***	12.2867***	-20.67***
NIKKEI	0.0003	0.0138	-0.3026	4.5760	2467.13***	17.0018***	-21.04***
SPTSX	0.0002	0.0096	-1.8155	25.757	78,343.5***	41.7380***	-10.93***
SPX	0.0005	0.0110	-0.7790	9.5335	10,805.2***	38.5681***	-13.55***
<u>RANGE</u>	Mean	Std. Dev.	Skew.	Ex. Kurt.	Jarque-Bera	Q(5)	ADF
CAC	0.0142	0.0089	2.4706	10.741	16,192.9***	3733.55***	-8.57***
GDAXI	0.0143	0.0089	2.4779	12.002	19,529.4***	3596.39***	-6.56***
FTSE	0.0126	0.0083	3.5583	24.289	74,203.1***	3931.79***	-8.16***
FTMIB	0.0180	0.0113	3.5887	28.654	101,073***	3382.78***	-8.61***
NIKKEI	0.0116	0.0084	5.1205	57.640	396,984***	1871.18***	-8.27***
SPTSX	0.0097	0.0070	3.9901	31.495	122,277***	4750.06***	-5.96***
SPX	0.0111	0.0081	2.8837	14.587	28,501.5***	4607.59***	-6.05***
<u>5-MIN RV</u>	Mean	Std. Dev.	Skew.	Ex. Kurt.	Jarque-Bera	Q(5)	ADF
CAC	0.0001	0.00017	9.9869	166.65	3.26E+06***	4818.41***	-8.10***
GDAXI	0.0001	0.00016	7.7792	91.942	1.007E+06***	5225.84***	-8.77***
FTSE	9.6E-05	0.00023	16.637	392.73	1.79E+07***	2035.20***	-10.3***
FTMIB	0.0001	0.00016	6.5410	66.618	533,886***	4674.88***	-6.32***
NIKKEI	7.5E-05	0.00015	11.574	197.14	4.56E+06***	1660.67***	-9.55***
SPTSX	5.5E-05	0.00013	14.468	308.73	1.11E+07***	2497.40***	-10.4***
SPX	7.9E-05	0.00019	10.893	167.64	3.31E+06***	4015.42***	-9.03***

Note: Asterisk *, **, and *** denote rejections of null hypothesis at 10%, 5%, and 1% significance levels, respectively. The null hypothesis of the third and fourth moments are “Skewness = 0” and “Excess Kurtosis = 3”.

Table 2

List of models (and indices) names and abbreviations.

HAR-RV	HETEROGENEOUS AUTOREGRESSIVE MODEL OF REALIZED VARIANCE	CAC	FRANCE CAC40
HAR-RV-ONV	HAR-RV-Overnight Volatility	DAX	Germany DAX30
HAR-RV-CC	HAR-RV-Close-to-Close Volatility, Brooks et al. (2000)	FTSE	United Kingdom FTSE100
HAR-RV-PK	HAR-RV-Parkinson, Parkinson (1980)	FTSEMIB	Italy FTSE MIB
HAR-RV-GK	HAR-RV-Garman-Klass, Garman and Klass (1980)	NIKKEI	Japan NIKKEI 225
HAR-RV-RS	HAR-RV-Rogers-Satchell, Rogers and Satchell (1991)	SPTSX	Canada S&P/TSX
HAR-RV-YZ	HAR-RV-Yang-Zhang, Yang and Zhang (2000)	SPX	United States S&P500

the overnight return improves the forecasts for CAC and DAX and is the preferred model. The inclusion of the overnight return is also the preferred model for SPX based on the QLIKE measure and is the second preferred model for MIB and the third for NIKKEI (QLIKE only). Thus, it is only for FTSE, that the forecast model that includes overnight volatility is ranked poorly. The close-to-close volatility measure reports some success across the different markets. For example, its inclusion leads to the preferred forecast model for MIB and NIKKEI, second preferred for FTSE and third for CAC and DAX. However, it is not preferred for SPX.

The inclusion of range estimators does not consistently receive support across the different markets. Although on an individual basis, there is some evidence that the Parkinson and Garman-Klass volatility estimators can improve forecasts over the plain HAR-RV model. For the Parkinson variable, this provides the preferred model for FTSE using the QLIKE measure. For the Garman-Klass variable, this provides the preferred forecasts for MIB (HMSE measure) and a top three forecast for FTSE (QLIKE measure), MIB (QLIKE measure) and NIKKEI (HMSE measure). This also means that the more involved range estimators, Roger-Satchell and Yang-Zhang, do not contain new information for forecasting realized volatility. Moreover, the HAR-RV model is not a preferred model for any market, although it does feature within the MCS several times.

5. Summary and conclusion

We examine whether range estimators, derived from OHLC prices, contain important information for the future realized volatility in G7 stock markets. Including close-to-close, overnight volatility, Parkinson, Garman-Klass, Roger-Satchell, and Yang-Zhang estimators as an exogenous variable in the HAR-RV model, we consider whether this improves volatility forecasts over a HAR-RV model using the QLIKE and HMSE metrics as well as the model confidence set approach that provides a statistical comparison of the different loss functions.

The results provide several noteworthy points. There is no single model that is preferred across all markets. However, the inclusion of the overnight return volatility is preferred for CAC and DAX and SPX on the QLIKE measure. It is also a top three preferred measure for MIB and NIKKEI. Only for FTSE, does the inclusion of overnight volatility not improve forecasts over the HAR-RV model. The addition of close-to-close volatility also provides notable forecast improvements for all series except the SPX. Moreover, it is preferred

Table 3

Rolling window forecasting results measured by QLIKE and HMSE and MCS procedure.

CAC	QLIKE	p-value	Rank	CAC	HMSE	p-value	Rank
HAR-RV	0.1675	eliminated	2	HAR-RV	0.7861	0.3078	3
HAR-RV-ONV	0.1517	1.0000	1	HAR-RV-ONV	0.7257	1.0000	1
HAR-RV-CC	0.1688	eliminated	3	HAR-RV-CC	0.8068	0.0782	6
HAR-RV-PK	0.1766	eliminated	6	HAR-RV-PK	0.8008	0.1878	5
HAR-RV-GK	0.1724	eliminated	5	HAR-RV-GK	0.8181	0.1964	4
HAR-RV-RS	0.1700	eliminated	4	HAR-RV-RS	0.7893	1.0000	2
HAR-RV-YZ	0.1817	eliminated	7	HAR-RV-YZ	0.8534	0.0378	7
DAX	QLIKE	p-value	Rank	DAX	HMSE	p-value	Rank
HAR-RV	0.1756	eliminated	2	HAR-RV	0.9157	eliminated	–
HAR-RV-ONV	0.1657	1.0000	1	HAR-RV-ONV	0.8607	1.0000	1
HAR-RV-CC	0.1771	eliminated	3	HAR-RV-CC	0.9355	eliminated	–
HAR-RV-PK	0.1830	eliminated	6	HAR-RV-PK	0.8990	eliminated	–
HAR-RV-GK	0.1849	eliminated	7	HAR-RV-GK	0.9134	eliminated	–
HAR-RV-RS	0.1790	eliminated	4	HAR-RV-RS	0.9216	eliminated	–
HAR-RV-YZ	0.1810	eliminated	5	HAR-RV-YZ	0.9602	eliminated	–
FTSE	QLIKE	p-value	Rank	FTSE	HMSE	p-value	Rank
HAR-RV	0.2851	0.7680	5	HAR-RV	2.8416	eliminated	–
HAR-RV-ONV	0.3140	eliminated	7	HAR-RV-ONV	2.8504	eliminated	–
HAR-RV-CC	0.2816	1.0000	2	HAR-RV-CC	2.7036	eliminated	–
HAR-RV-PK	0.2806	1.0000	1	HAR-RV-PK	2.5594	1.0000	1
HAR-RV-GK	0.2887	1.0000	3	HAR-RV-GK	2.6977	eliminated	–
HAR-RV-RS	0.2892	0.4778	6	HAR-RV-RS	2.9010	eliminated	–
HAR-RV-YZ	0.2860	1.0000	4	HAR-RV-YZ	2.8410	eliminated	–
FTSEMIB	QLIKE	p-value	Rank	FTSEMIB	HMSE	p-value	Rank
HAR-RV	0.1509	0.5504	4	HAR-RV	1.6260	0.5534	6
HAR-RV-ONV	0.1492	1.0000	2	HAR-RV-ONV	1.5580	0.4120	2
HAR-RV-CC	0.1470	1.0000	1	HAR-RV-CC	1.5920	1.0000	4
HAR-RV-PK	0.1518	0.1598	6	HAR-RV-PK	1.6090	eliminated	–
HAR-RV-GK	0.1519	1.0000	3	HAR-RV-GK	1.4940	1.0000	1
HAR-RV-RS	0.1583	eliminated	7	HAR-RV-RS	1.6650	1.0000	5
HAR-RV-YZ	0.1504	0.3996	5	HAR-RV-YZ	1.5570	0.2766	3
NIKKEI	QLIKE	p-value	Rank	NIKKEI	HMSE	p-value	Rank
HAR-RV	0.2863	0.9072	2	HAR-RV	1.6990	0.1100	6
HAR-RV-ONV	0.2903	0.7524	3	HAR-RV-ONV	1.7850	0.0312	7
HAR-RV-CC	0.2835	1.0000	1	HAR-RV-CC	1.6020	1.0000	1
HAR-RV-PK	0.5478	eliminated	7	HAR-RV-PK	1.7560	1.0000	4
HAR-RV-GK	0.4080	eliminated	6	HAR-RV-GK	1.7240	1.0000	3
HAR-RV-RS	0.3754	eliminated	5	HAR-RV-RS	1.7870	1.0000	5
HAR-RV-YZ	0.2877	0.6694	4	HAR-RV-YZ	1.6280	1.0000	2
SPTSX	QLIKE	p-value	Rank	SPTSX	HMSE	p-value	Rank
HAR-RV	0.2532	eliminated	5	HAR-RV	1.8270	eliminated	–
HAR-RV-ONV	0.2175	1.0000	1	HAR-RV-ONV	1.7780	eliminated	–
HAR-RV-CC	0.2729	eliminated	7	HAR-RV-CC	1.7360	eliminated	–
HAR-RV-PK	0.2405	eliminated	2	HAR-RV-PK	1.4420	1.0000	1
HAR-RV-GK	0.2517	eliminated	4	HAR-RV-GK	1.5550	eliminated	–
HAR-RV-RS	0.2586	eliminated	6	HAR-RV-RS	1.7160	eliminated	–
HAR-RV-YZ	0.2480	eliminated	3	HAR-RV-YZ	1.7200	eliminated	–
SPX	QLIKE	p-value	Rank	SPX	HMSE	p-value	Rank
HAR-RV	0.3012	0.0070	5	HAR-RV	3.2730	eliminated	–
HAR-RV-ONV	0.2942	1.0000	3	HAR-RV-ONV	3.4880	eliminated	–
HAR-RV-CC	0.4054	1.0000	7	HAR-RV-CC	3.1940	eliminated	–
HAR-RV-PK	0.3334	1.0000	6	HAR-RV-PK	1.9250	1.0000	1
HAR-RV-GK	0.2908	1.0000	1	HAR-RV-GK	2.5470	eliminated	–
HAR-RV-RS	0.2993	0.0016	4	HAR-RV-RS	3.0710	eliminated	–
HAR-RV-YZ	0.2934	1.0000	2	HAR-RV-YZ	2.9860	eliminated	–

Note: To save more space, loss functions and MCS procedure results are merged in this table. QLIKE and HMSE values are obtained by one-step-ahead rolling window forecasting method. Window size is 600 observations that is used as in-sample estimation. The out-of-sample consists of 2180 observations. P-value and rank results are received from the MCS procedure.

on the QLIKE measure for MIB and NIKKEI. Two of the more simpler range methods, the Parkinson and Garman-Klass, also provide some forecast success for some of the markets, i.e., the forecast model is top three in performance for at least one forecast measure for FTSE, MIB, and NIKKEI. The HAR-RV model itself, however, does not provide the preferred forecast for any market.

The HAR-RV model has become the dominant volatility forecasting model. The results of this paper highlight that the inclusion of additional variables can improve that forecasting performance. Moreover, it is the relatively simple overnight volatility measure that consistently provides such an improvement for all but one market. Volatility forecasts are key ingredients into portfolio management, market timing strategies and derivative pricing, among other things, it is hope that the results here will be of use to investors as well as

academics.

CRedit authorship contribution statement

Burak Korkusuz: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Dimos Kambouroudis:** Conceptualization, Methodology, Resources, Supervision, Writing – review & editing. **David G. McMillan:** Conceptualization, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – review & editing.

Data availability

The authors do not have permission to share data.

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